Study of Mobile Ad hoc Network routing protocols based on NS3 and SUMO simulation

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1. Introduction

Mobile Ad Hoc Networks (MANETs) are autonomous systems of mobile devices connected by wireless links without any pre-existing infrastructure such as routers or access points. They are decentralized and dynamic networks where each device is free to move independently in any direction and organize itself arbitrarily. Thus, the network's wireless topology may change rapidly and unpredictably. Furthermore, every node in the network functions as a router, forwarding data packets for other nodes. As such, they communicate thanks to multi-hop routing, where packets are passed from one node to another until they reach their destination.

The primary characteristic of MANETs is their self-configuring ability. The network's topology is dynamic due to the constant movement of nodes, making it challenging yet crucial to maintain robust communication. This nature of MANETs makes them suitable for situations where temporary network setups are needed or where it is impractical to rely on fixed infrastructure, such as in disaster areas, battlefield scenarios, for robotics or for large-scale outdoor setup.

As such, one of the critical challenges in MANETs is the design of efficient routing protocols that can handle the high degree of network topology variability. They must be adaptive to efficiently find and maintain routes between nodes under the conditions of changing topology and possibly variable wireless link quality. Among the numerous protocols developed for these networks, the Ad hoc On-Demand Distance Vector (AODV) and the Dynamic Source Routing (DSR) protocols stand out due to their unique approaches to data packet routing. Other challenges include, but are not limited to, security, power consumption, and bandwidth optimization.

This report aims to conduct an in-depth performance comparison of the AODV and DSR routing protocols within MANETs by employing realistic mobility patterns generated by the Simulation of Urban MObility (SUMO) package. SUMO is an open source and continuous road traffic simulator designed to handle large road networks. By using these real-world vehicle mobility traces, this study seeks to assess how well each routing protocol performs in environments that closely mimic real-world vehicle movements. Key performance metrics to be evaluated include packet delivery ratio, network throughput, end-to-end delay and routing overhead under different mobility scenarios. This realistic approach will provide valuable insights into the practical applications and limitations of each routing protocol, thereby guiding optimal protocol selection for specific MANET deployments.

2. Background

A - Fundamentals of Routing Protocols

Routing protocols for MANETs can be classified into two categories based on their approach to route discovery and maintenance: proactive and reactive protocols. These categories represent different methodologies for handling information about the paths through the network.

Proactive Routing Protocols, also known as table-driven routing protocols, maintain fresh lists of destinations and their routes by periodically distributing routing tables throughout the network. Common examples of proactive protocols include DSDV (Destination-Sequenced Distance Vector) and OLSR (Optimized Link State Routing). They may face some scalability issues as the network increases in size due to the routing tables becoming too large.

On the other hand, Reactive Routing Protocols, known as on-demand routing protocols, create routes only when desired by the source node. When a route to a new destination is required, the protocol initiates a route discovery process within the network. Examples include AODV (Ad hoc On-Demand Distance Vector) and DSR (Dynamic Source Routing). Compared to proactive routing protocols, they notably offer reduced overhead but suffer from greater delay when sending packets.

Hybrid Routing Protocols also exist and try to combine the advantages of both types of protocol.l Most of the time, they will distinguish close vicinities of nodes and use specific protocols inside and between them depending on the characteristics of the zones. The most common ones are ZRP (Zone routing protocol) and TORA (Temporarily Ordered Routing Algorithm).

B - DSR Protocol

As explained previously, Dynamic Source Routing (DSR) is a reactive protocol designed for Mobile Ad Hoc Networks (MANETs). It relies on source routing, where the entire path from source to destination is included in the packet's header. This approach allows intermediate nodes to forward packets without needing extensive routing tables, reducing overhead in the process.

Key Features in DSR :

1. Source Routing: The source node specifies the complete route to the destination in the packet's header. This approach reduces the need for intermediate nodes to maintain routing information.
2. Route Cache: DSR uses a route cache to store previously discovered routes. This cache helps reduce overhead by avoiding repeated route discoveries for frequently used routes.

3. On-Demand Routing: DSR establishes routes only when a data packet needs to be sent, minimizing routing overhead in scenarios where traffic is sporadic or the network topology changes frequently.

In DSR, when a source node requires a route to a destination, it broadcasts a Route Request (RREQ) packet to its neighboring nodes. The RREQ contains the source and destination addresses, as well as an unique identifier allowing nodes to discard already received requests. Each node that forwards the RREQ appends its identifier to the packet's header, creating a path back to the source. If the destination or an intermediate node has a valid route, a Route Reply (RREP) is generated and sent back either along the reverse path or, if available, through a cached route from the destination node to the source. The reverse path is stored in each node's route cache, allowing them to reuse it in the future, thereby reducing the need for frequent route discoveries.

Route maintenance in DSR involves detecting broken links and removing invalid routes from the route cache. If a node detects a broken link, it sends a Route Error (RERR) message to inform others, triggering a new route discovery if no other route is found. This mechanism is only activated when a source node needs to send data packet DSR's use of route caching and source routing minimizes the need for periodic routing updates, reducing overhead. However, this approach can lead to scalability issues in larger networks and larger packet headers due to source routing

C - AODV Protocol

The Ad hoc On-Demand Distance Vector (AODV) protocol is another reactive routing protocol used in Mobile Ad Hoc Networks (MANETs) and other wireless ad hoc networks. It extends the principle behind the DSR protocol and tries to remove the disadvantage of source routing.

Key Features of AODV:

1. Hop-by-Hop Routing: AODV utilizes hop-by-hop routing, meaning each node along the route forwards data packets to the next hop as defined in the route entry cached in its routing table. The packet header does not contain the route to its destination.
2. Sequence Numbers: To ensure the freshness of routing information and to prevent routing loops, AODV uses sequence numbers that are updated as the network topology changes. Each node maintains its own sequence number, and each route table entry also includes the destination node’s sequence number.
3. Path Setup and Maintenance: AODV establishes routes that are only needed by source nodes. This reduces the overhead and bandwidth consumption, making AODV scalable in larger networks with many nodes.
4. Route Discovery: When a node needs to send data to a destination for which it does not have a route, it initiates a route discovery process. This is done by broadcasting a Route Request (RREQ) packet to its neighbors and receiving a Route Reply (RREP) in a similar fashion as with DSR.

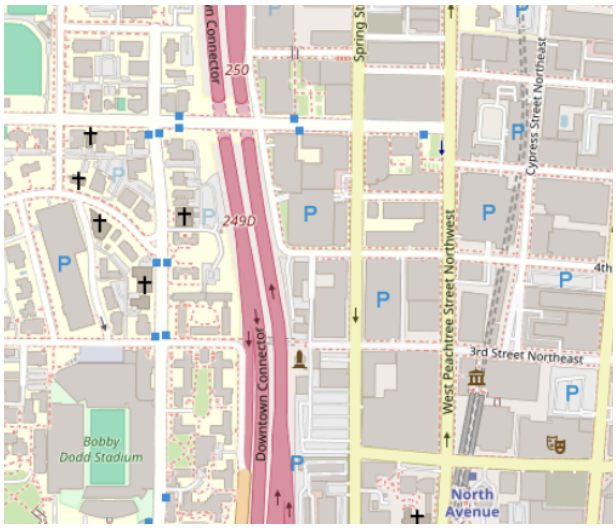
5. Route Maintenance: Once a route is established, it is maintained by route maintenance procedures. If a link break in an active route is detected, a Route Error (RERR) packet is sent to the affected upstream nodes. They will update their routing table and if needed, launch a new route discovery procedure when sending a data packet.

AODV is favored in environments where the network topology changes frequently. It greatly reduces the header size and proposes a quick route construction. However, it needs much more control messages compared to DSR and has more risk of routing loops.

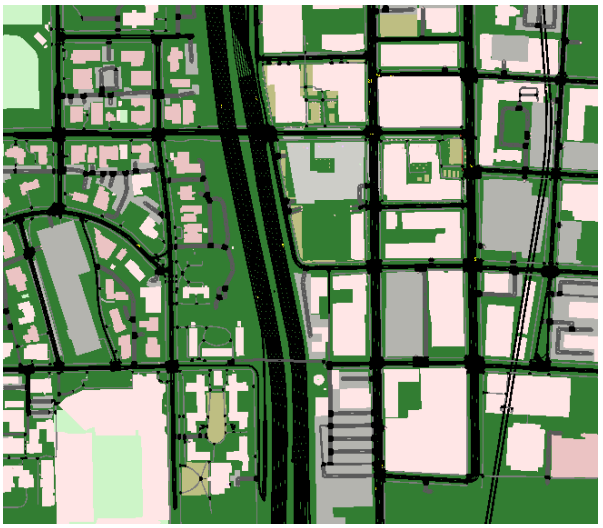
3. Methodology

A - Traffic Simulation Tools

SUMO (Simulation of Urban MObility) is an open-source traffic framework designed to model urban traffic, including vehicles, pedestrians, and public transport. Thanks to it, we could easily integrate realistic mobility patterns for the nodes in the ns3 simulation. We used it in conjunction with OSMWebWizard, a tool that allows us to generate our SUMO traffic simulations based on real world data from the OpenStreetMap map.

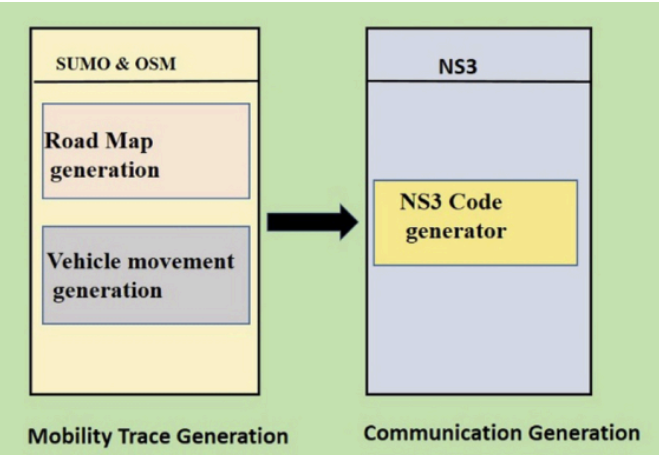


OSMWebWizard map used for the simulation
Part of Georgia Tech



Map generated with the SUMO trace

When generating a SUMO simulation, it creates a trace including all vehicle movements. We can then convert it into a suitable format to be read by our ns3 simulation. This latter will be able to parse it to create as many nodes as the number of vehicles. It will also be able to know their initial position, their travel, as well as their entry and exit time in the simulation.



B - Network Configurations and Scenario Design

In order to design our simulation in ns3, we decided to start from the file *manet-routing-compare.cc,* an example of the framework allowing us to compare routing protocols. We added the support to read SUMO trace files thanks to the provided code and changed a few things, notably the mobility model and the statistics collection.

The code used by the two different protocols is exactly the same, as the only difference is the type of routing helper used and its setup. We decided to use the OnOff application, as it is widely used in network research to study how networks perform under different conditions and examine QoS mechanisms. It provides several advantages, mainly due to the bursty traffic patterns it can generate, as it can alternate between periods of active transmission, *on*, and inactivity, *off*.

The WiFi is set in ad hoc mode, a type of local area network (LAN) that is built spontaneously to enable two or more wireless devices to be connected to each other without requiring typical network infrastructure equipment, such as a wireless router or access point. It is also configured with the Friis loss model, in order to model the path loss incurred in a free space environment, devoid of any objects that create absorption, diffraction, reflections, or any other characteristic-altering phenomenon.

We set up the applications generating the data packets in such a way: when a node enters the network, i.e. it enters the simulation as defined in the SUMO trace file, it chooses randomly another active node in the network. It will then start broadcasting packets until one of them leaves the network.

C - Parameters for Performance Evaluation

We decided to use the following parameter for all our simulations:

- Packet size: 512 bytes
- Rate: 2048 bps
- Transmit signal power: 10 dBm
- Simulation duration: 600s

In the end, we set up the On Off application to simply broadcast constantly as we wanted to observe the behavior of our network and the impact of a saturated network. We also decided to use the IEEE 802.11p standard for the Wifi, as it is designed to support wireless access in vehicular environment (WAVE). It is adapted to our use case as its goal is to facilitate data exchange between high-speed vehicles.

In order to compare the two protocols and how they perform, we did multiple simulations with varying the input mobility trace. The main difference between all trace files is the increasing number of nodes, which therefore increase the number of packets generated as the number of sources and sinks increase.

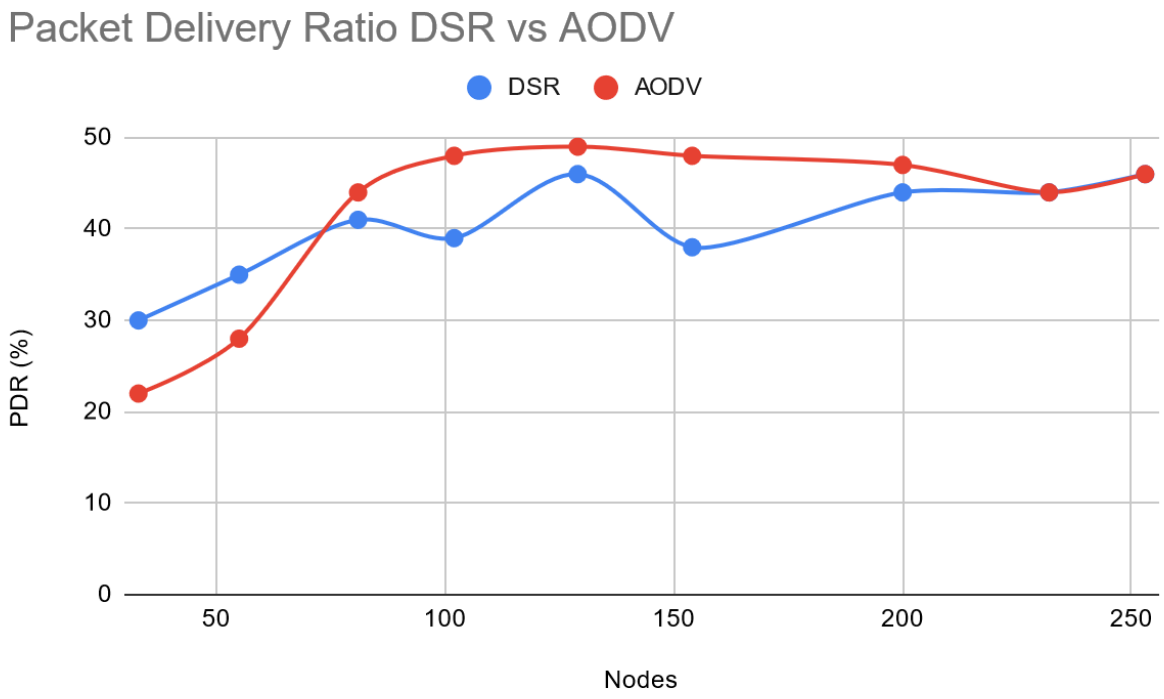
We assessed the following performance :

- The Packet Delivery Ratio (PDR) measures the proportion of successfully delivered data packets relative to the total packets sent. A high PDR indicates a reliable network, while a low PDR suggests packet loss, possibly due to congestion or routing issues.
- The Network Throughput is the rate at which data is successfully transmitted across a network. It gives an idea of the network's capacity in terms of bandwidth.
- The Average End-to-End Delay measures the average time it takes for a data packet to travel from the source to the destination across a network. It includes all possible delays caused by buffering, transmission, propagation, processing, and queuing. A lower average end-to-end delay indicates a more efficient network, while a high one may cause issues when dealing with some applications.
- The overhead ratio quantifies the bandwidth used by control packets for maintaining routes and network topology information in comparison to the bandwidth used for actual data transmission. It reflects the efficiency of the routing protocol in terms of bandwidth consumption. A lower overhead ratio means that a greater portion of the network’s capacity is being used for transmitting user data, rather than routing information.

4. Performance Evaluation

A - Packet Delivery Ratio

This graph compares the Packet Delivery Ratio (PDR) of the DSR and AODV routing protocols as the number of nodes in a network increases.



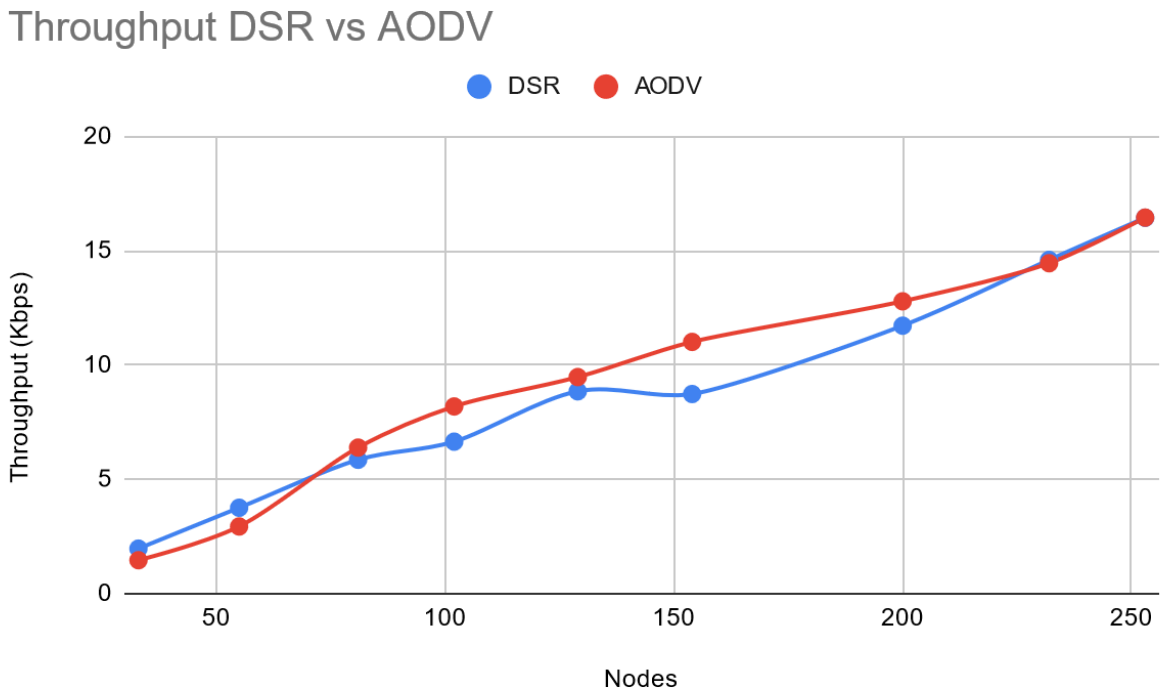
- DSR Over AODV in PDR: The DSR protocol starts with a lower PDR but surpasses AODV at around the 70-node mark and maintains a higher or similar PDR compared to AODV for the remainder of the node range. However, they converge to the same values for a high number of nodes, which is probably due to the limits of our chosen network transmission model.
- PDR Variation with Node Increase: Initially, both protocols improve in PDR as the number of nodes increases up to around 100 nodes. However, after this point, PDR variations are less consistent, with AODVstabilizing before showing a slight decline, while DSR shows a lot of fluctuation.
- AODV Stability: AODV shows a more stable evolution after 100 nodes with less fluctuation in PDR, suggesting it may be more consistent in delivery performance in larger networks compared to DSR.

From this graph, we can conclude that while DSR may offer a higher PDR for a small enough network size, AODV exhibits a higher and more stable PDR for a bigger network. From a certain point, DSR suffers due to its reliance on route caching, which can lead to outdated routes if the cache isn't frequently updated. As such,

each protocol's performance in terms of PDR might be better suited to different network sizes and requirements. The similarity of the final PDR for the two protocols rather suggests the limits of our simulation

B - Network Throughput

The graph compares the throughput of the DSR and AODV routing protocols with respect to the number of nodes in a network. Throughput is measured in Kbps (kilobits per second).

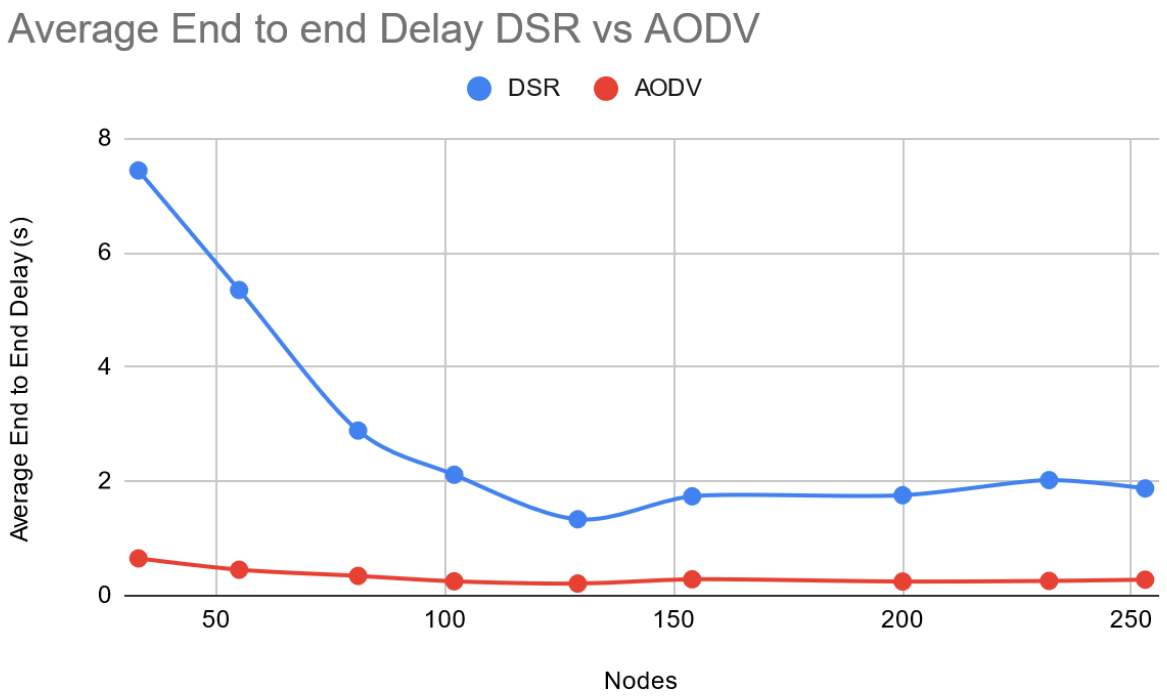


- General Trend: Both protocols exhibit an upward trend in throughput as the number of nodes increases. This suggests that our model does not exploit the MANET protocols up to their limit as no bottleneck phenomenon happens.
- AODV Outperforms DSR: From 70 nodes upwards, as in the PDR graph, the AODV protocol has a higher throughput than the DSR protocol. We also observe their respective throughput to converge when reaching 230 nodes.

Overall, the graph indicates that AODV might be more effective in terms of throughput in larger networks compared to DSR, particularly as the network size increases.

C - Average End to End Delay

The graph compares the average end-to-end delay of packets in seconds using the DSR and AODV routing protocols with respect to the number of nodes in a network.

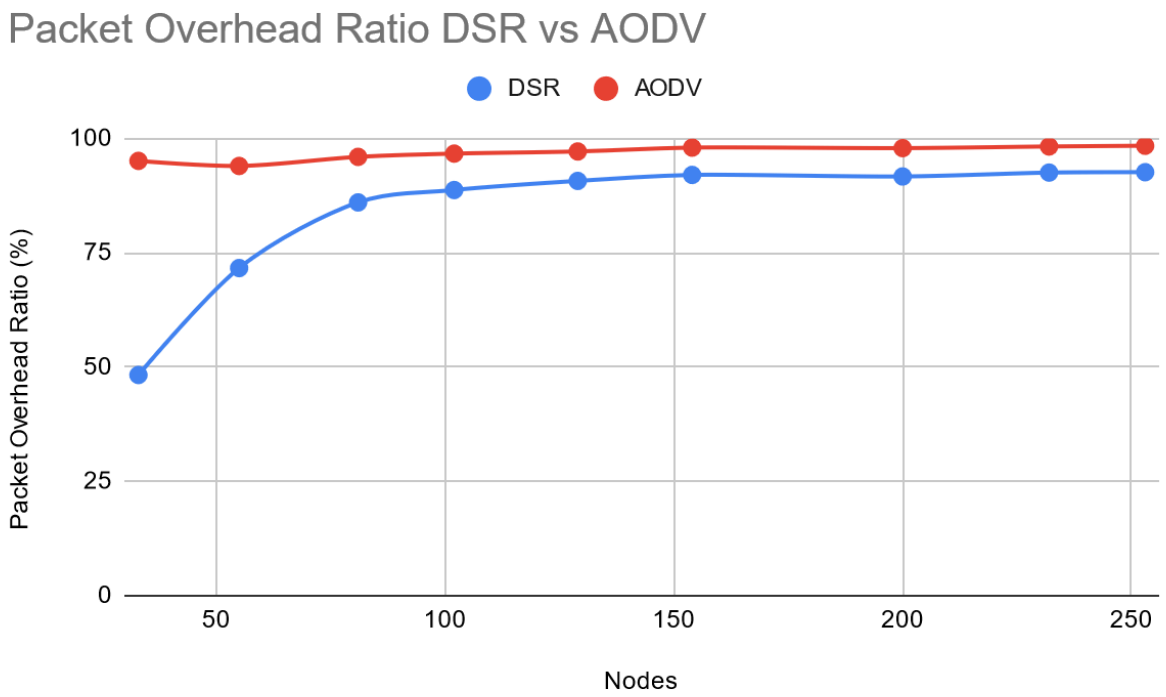


- DSR Evolution: As the number of nodes increases up to 130, we observe that the average end-to-end delay for the standard network decreases significantly from 7.4 to 1.3s. This suggests that the network becomes more efficient as it scales up to a certain point, probably due to more link breaking when the network is less dense.
- AODV Evolution: In contrast to DSR, the average end-to-end delay for the AODV protocol remains much lower and more stable. A very slight decrease also happens until we reach 100 nodes.
- General Trend for lots of Nodes: We also observe a slight increase in delay for both protocols as the number of nodes goes starting between 150 and 200 nodes, which could indicate a point where adding more nodes begins to have a negative impact on the network's performance, even though it is very low.

In the end, the AODV protocol maintains a much lower and more stable delay across all node counts, suggesting it is much more efficient when considering this metric.

D - Packet Overhead Ratio

The following graph compares the ratio of non-data packets, such as RREQ, RREP or RRER packets, over the total number of packets transmitted during the simulation.



- DSR Evolution: The overhead for DSR starts slightly below 50% for 30 nodes and shows a quick increase as the number of nodes grows, reaching 88% at 100 nodes. After that, its increase becomes quite slight as it only reaches 92.5% for 250 nodes. This shows that as the network scales up, the DSR protocol requires more and more control packets to maintain its routes.
- AODV Evolution: The overhead for AODV stays much more stable but is also much higher as it begins at 95% for 30 nodes. As the network size increases, there is a slight increase reaching 98.5% at 250 nodes. No matter the size of the network, AODV requires approximately the same number of control packets when transmitting data to maintain its routes.

A high overhead ratio can impact the network's performance, as more bandwidth is consumed by control packets, leaving less available for actual data transmission. The data suggests that DSR offers much better performance for small networks, but as the number of nodes increases, its performance becomes more similar to the ones of AODV. In the end, DSR still remains a better choice than AODV as it requires 6% less control packets.

5. Conclusion and Future Work

A - Summary of Key Findings

Dynamic Source Routing (DSR) can be effective in small mobile networks, where it offers better packet delivery ratio and throughput compared to AODV, as well as fewer routing overhead. However, its end-to-end delay is much higher as the node cache becomes quickly outdated and frequent route errors trigger additional route discoveries. On the other end, Ad hoc On-Demand Distance Vector (AODV) generally shows superior performance in large networks. It achieves higher packet delivery ratios (PDR), higher throughput and much lower end-to-end delay, even though its routing overhead is quite high, as it better adapts to changing network conditions. Overall, both AODV and DSR offer benefits in terms of efficiency and scalability, with each protocol showing strengths and weaknesses depending on the network context but AODV seems to offer better scalability capabilities in a mobile and dynamic network.

B - Improvements and Future Research Directions

Although our study allows us to compare the 2 main reactive MANETs protocols (AODV and DSR) in terms of Packet Delivery Ratio and Throughput, it might be insufficient to properly choose between those protocols, as suggest the similar result in terms of packet delivery ratio and throughput for a very high number of nodes. A more realistic study should try different packet generation models, study the impact of a more bursty traffic and also observe the impact of speed variations. It could also add more complex propagation models which include buildings instead of the Frii's free space model we are using, for example.

Furthermore, here are some suggestions for future research directions in this domain:

- We could focus on developing and studying new routing protocols or improving existing ones to better handle the increasing complexity of MANETs. This could include Adaptive Routing, Energy-Efficient Routing, Hybrid Routing.
- Security and privacy are a significant concern in MANETs as most of them do not implement any effective access control. Future research could explore intrusion detection and prevention, secure routing or privacy-preserving communication.
- We could also try to integrate MANETs with some emerging technologies in the context of the Internet of Things (IoT), edge computing or autonomous vehicles.

Appendix

In order to run the simulation, we need to run the *manet_routing_simulation.cc* file and specify the two following parameters:

- *protocol*, which can be set as AODV or DSR. The code also supports OLSR and DSDV, even though we did not use them.
- *SUMOtraceFile*, which specifies the SUMO mobility trace to use. To generate it, we need first to generate the trace in a *.xml* file from the SUMO scenario and then convert it into the *.tcl* format suitable for ns3.

References

Kanthe, A.M.; Simunic, D.; Prasad, R. “Comparison of AODV and DSR on-demand routing protocols in mobile ad hoc networks”. In Proceedings of the 2012 1st International Conference on Emerging Technology Trends in Electronics, Communication & Networking, Surat, India, 19–21 December 2012; pp. 1–5.

M. M. Abdellatif and O. Aly, “Studying Routing Issues in Vanets UsingNs-3 and SUMO” Journal of Communications, vol. 17, no. 9, 2022

S.-S. Kang, Y.-E. Chae and S. Yeon, “Vanet routing algorithm performance comparison using ns-3 and sumo”, *2017 4th International Conference on Computer Applications and Information Processing Technology (CAIPT)*, pp. 1-5, 2017.

Sarao, P. “Comparison of AODV, DSR, and DSDV routing protocols in a wireless network.” Journal of Communications, 13(4), 175–181, 2018

“Types of MANET in Computer Network.” GeeksforGeeks. Accessed May 1, 2024. https://www.geeksforgeeks.org/types-of-manet-in-computer-network/